



Aboveground biomass equations for the predominant conifer species of the Inland Northwest USA

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ABSTRACT

Assessment of aboveground biomass stocks in the coniferous forests of the inland northwest USA is important for timber, bioenergy, and carbon inventories, as well as for wildfire risk determination. In this study, individual tree biomass equation systems are developed for 7 regionally important conifer species using data from 470 felled trees sampled across 84 stands and spanning a range of diameters at breast height (1.37 m; dbh) running from 5 cm to 105 cm. The equation systems permit estimation of crown biomass components (i.e., foliage, dead branches, and live branches by size class) and stem components (above-stump stemwood and stem bark), as well as compatible estimates of (sub)totals. The systems draw on commonly collected inventory variables including dbh, tree height, and live crown length. All biomass components scaled approximately linearly with dbh on the logarithmic scale, but equation systems drawing on both dbh and height provided more accurate estimates for all species; systems drawing additionally on live crown length provided more accurate estimates still for all species but one. In line with previous work, incorporation of live crown length improved live crown component equations most, but also improved stem component equations for two species. Across species and systems, stem components and subtotals were most accurately estimated (mean absolute errors ~10%) while dead branch biomass estimation proved least tractable (mean absolute errors >50%). Overall, the reported biomass equation systems draw on the largest felled tree samples collected from the region, and provide the most comprehensive basis developed to date for regional forest biomass assessments over the inland northwest.

1. Introduction

The temperate coniferous forests of the inland northwest, a mountainous region of the USA encompassing parts of eastern Washington, northern Idaho, and western Montana, span a broad range of elevations and diverse climatic regimes. In the last century, management of this landscape was framed primarily by timber harvesting and wildfire suppression. These remain predominant activities across the region, though today fuels mitigation, bioenergy extraction, and carbon sequestration are also important forest management objectives. Crucial to the effectiveness and sustainability of all these activities is the ability to reliably assess forest biomass stocks. Biomass is of central interest in fuels and bioenergy assessments, has direct linkages to forest carbon inventory, and is increasingly being used as a basis for the trading of merchantable wood products. Despite this, there are presently no comprehensive sets of aboveground biomass equations for the major conifer species of the region. The purpose of this research is to advance a set of such equations calibrated from regional felled tree data.

The most widespread and commercially important coniferous species in the inland northwest are Douglas-fir (*Pseudotsuga menziesii* var.

glauca; PSME), lodgepole pine (*Pinus contorta*; PICO), ponderosa pine (*Pinus ponderosa*; PIPO), western larch (*Larix occidentalis*; LAOC), grand fir (*Abies grandis*; ABGR), subalpine fir (*Abies lasiocarpa*; ABLA), and Engelmann spruce (*Picea engelmannii*; PIEN). An extensive study of the biomass allometries of these 7 species (and 4 additional species) was conducted by Brown (1978). However, his study focused on crown biomass – no data were collected on stem components for trees with diameter at breast-height (dbh) above 10 cm. This focus was owing to the primary role of crown components in wildfire dynamics and to the fact that wood products in the inland northwest were traded almost exclusively on a volumetric basis at that time. Revisions to Brown's foliar biomass equations were made by Moeur (1981), and adjustments for a subset of the crown equations were proposed by Gray and Reinhardt (2003). Nonetheless, Brown's equation systems remain the standard basis for tree-level biomass determination in regional studies.

For analyses of a national or broader scope, the equations developed by Jenkins et al. (2003) (later revised by Chojnacky et al., 2014) have been applied to inventory data from inland northwest forests. Indeed, the USA's Forest Inventory & Analysis (FIA) program utilizes the former equations across the western USA, albeit only as a method for

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distributing biomass among tree components (Woodall et al., 2011). As Jenkins et al. emphasized, these equations were generalized in order to “provide a consistent basis for evaluating forest biomass across regional boundaries” (p. 13). The generalizations, including the aggregation of species into groups (e.g., all *Pinus* species for total mass determination; all conifer species for component biomass distribution) and the use of dbh as the sole predictor of biomass, ought to serve well in capturing overall trends and ensuring that estimates do not vary as a function of artificial boundaries (e.g., state lines). Yet these same generalizations also mean that the equations cannot account for variations in height-diameter ratios, crown ratios, or even species in many cases. Such variations are common within a region as diverse in growing conditions and management practices as the inland northwest.

Needed within the region are biomass equations that can accommodate these sources of variation and provide whole-tree and component estimates calibrated to the species and predominant growing conditions of the inland northwest. As shown by Case and Hall (2008), regionally-calibrated equations can provide biomass estimates with less bias and improved precision relative to generalized national-level equations, and with accuracy approaching that of localized equations. This can be achieved in part by confining the calibration data to the region (and species) of interest, but also by expanding the suite of predictor variables to include others commonly collected in regional inventories. In the inland northwest, height-diameter ratios can vary substantially as a function of stand density, and thus height is commonly recognized as an important variable for regional stem volume estimation (see e.g. Table 3 in Woodall et al., 2011). Likewise, live crown length can vary with stand density and development, and both empirical studies (Brown, 1978; Evert, 1985) and allometric scaling theory (Mäkelä and Valentine, 2006) support its use as an important predictor of crown biomass.

The primary objective of this research was to develop systems of biomass equations for the aboveground components of the major conifer species in the inland northwest, and to calibrate the systems using regional felled tree data. The systems are intended for use in forest inventory, and thus focus on trends in biomass distributions with respect to commonly measured inventory variables (dbh, total height, live crown length, and derived variables). A secondary objective was to compare the performance of biomass equation systems employing distinct subsets of tree-level predictor variables – specifically, systems employing only dbh and systems employing only dbh and height.

2. Materials & methodology

2.1. Stand and tree selection

Felled tree data were collected during the summers of 2009–2015. Collection protocols varied across years owing to evolving objectives and constraints, but followed the same general outline. Throughout the period, study stands were selected opportunistically across federal, state, tribal, and private lands with the aim of securing a sample of trees having broad geographic, climatic, and size distributions. Study stands were not confined to active or planned timber sales, but neither were such stands excluded; only stands thinned or treated within 5 years were such stands excluded.

Within selected stands, candidate sample trees were identified at one or more sample points. In most stands, sample points were located systematically on the universal transverse Mercator (UTM) grid at 50 m intervals, with points located no closer than 25 m to a road. In other stands, particularly active logging sites, points were located opportunistically, though still with a 50 m offset and 25 m road buffer. From 2009 to 2012, candidate sample trees were identified at each sample point using a 2.3 m²/ha factor prism; up to two trees at each point were then selected for destructive biomass sampling. Final tree selection was generally made at random, but in some instances trees were selected purposively to broaden the species and size class distribution of the

sample. From 2013 onward, only the latter strategy was used and up to 4 trees were purposively selected within the vicinity of a sample point. In all years, trees were selected for sampling only if they exhibited (i) dbh ≥ 5 cm; (ii) no obvious forking or broken/dead tops; and (iii) no obvious insect or disease damage. Once selected, sample trees were directionally felled so as to minimize branch entanglement and breakage. Total height and height to the base of the live crown were then measured. The base of the live crown was defined as the lowest point on the stem where at least two live branches separated by an angle of $\geq 90^\circ$ were attached (USDA Forest Service, 2009, p. 42).

2.2. Estimation of stem biomass

The stem of a sample tree was defined as the largest diameter limb emanating from any forkings between a stump height of 30 cm from ground and an upper-stem diameter of 5 cm. All branches attached to the stem as well as the tree tip above an outside-bark diameter of 5 cm were treated as crown materials. Inside and outside bark diameter measurements were made along the length of each stem, and three or more discs were then cut between the ground line and the 5 cm top. Discs were drawn systematically with a random start except in a few stands where merchandizing restrictions required that the discs be taken at prescribed log lengths. Also, in 2013, felled trees were cut into segments at 0.15 m, 1.37 m, 2.44 m, and then every 2.44 m up the stem to the 5 cm top. The segments were weighed green in the field (wood and bark combined) before discs were extracted from the top or bottom. Discs were weighed and measured in the field, then weighed again after oven-drying at 105 °C to a constant mass. A detailed description of how the disc and stem measurements were combined to obtain individual tree stemwood and stembark biomass estimates (m_w and m_b , respectively; see Table 1) is provided in the [supplementary materials](#) (p. S1).

2.3. Estimation of crown biomass

Live crown biomass was assessed using aggregated randomized branch sampling (RBS) techniques (Schlecht and Affleck, 2014; Gregoire and Valentine, 2008). Between 5 and 10 live branches were selected from each sample tree by RBS, with the target number of branches set as an increasing function of tree dbh. Conditional RBS branch selection probabilities were set proportional to branch basal area. Selected live branches were separated into fuel time-lag classes based on diameter thresholds: diameter ≤ 0.635 cm (1 h fuels); 0.635 cm

Table 1
Table of symbols.

Symbol	Description
d	Outside-bark diameter at breast height (1.37 m; cm)
h	Total tree height (m)
l	Live crown length (m)
m_d	Dry mass of dead first-order branches (kg)
m_f	Dry mass of foliage (kg)
m_{l1}	Dry mass of materials ≤ 0.635 cm in diameter in live branches (kg)
m_{l10}	Dry mass of materials 0.635–2.54 cm in diameter in live branches (kg)
m_{l100+}	Dry mass of materials >2.54 cm in diameter in live branches (kg)
m_l	Dry mass of live first-order branches ($m_{l1} + m_{l10} + m_{l100+}$; kg)
m_c	Dry mass of the crown ($m_d + m_f + m_l$; kg)
m_w	Dry mass of wood in the main stem from a 30 cm stump to a 5 cm top (kg)
m_b	Dry mass of bark in the main stem from a 30 cm stump to a 5 cm top (kg)
m_s	Dry mass of the main stem from a 30 cm stump to a 5 cm top ($m_w + m_b$; kg)
m_a	Abovestump dry mass of the tree based on a 30 cm stump ($m_c + m_s$; kg)
$c_{dh,i}$	Principal component i of $\ln d$ and $\ln h$
$c_{dhl,i}$	Principal component i of $\ln d$, $\ln h$, and $\ln l$
β_{ci}	Parameter of the expectation function of biomass component c
α_i	Parameter of the live branch disaggregation function
θ_{ci}	Parameter of the variance function of biomass component c

< diameter ≤ 2.54 cm (10 h fuels); diameter > 2.54 cm (100+ hour fuels). Material was then stored in paper bags for processing in the laboratory.

In general, RBS was applied only to first-order branches > 0.635 cm in diameter. Epicormic and small live branches (basal diameter ≤ 0.635 cm) were treated as a separate stratum. The latter were weighed green in aggregate in intervals along the stem, with samples taken from each batch and bagged for dry mass determination. First-order dead branches were also removed and weighed in aggregate in intervals along the stem. Samples were taken from each batch, weighed green, and then stored in paper bags for oven drying and moisture content determination.

Foliage was removed from live branch components in the laboratory, but branch bark and wood were not separated. All crown materials were oven dried at 105°C to constant mass. The estimators with which crown measurements and branch samples were used to obtain individual tree crown component biomass estimates are detailed in the [supplementary materials](#) (p. S1).

2.4. Assimilation of additional biomass data

In addition to the data collected using the protocols described above, previously compiled tree biomass data from the inland north-west were available from [Brown \(1978\)](#). [Brown \(1978\)](#) collected crown biomass data for the 7 species of interest from 14 sites spread across northern Idaho and western Montana. Tree selection and measurement protocols differed from the procedures described above, but crown biomass was ultimately divided into the same components, including foliage (m_f), first-order dead branches (m_d), and live branch material by size class (m_{1h} , m_{10h} , and m_{100+}). One important difference was that [Brown](#) did not include the mass of the stem tip in those components. However, he did report tip dimensions and tip dry mass for each tree. Using this information, and assuming a conical form for the tip, the reported tip mass for each tree was proportionally reduced to a 5 cm basal diameter basis and fractionated into 1-h, 10-h, and ≥ 100 -h size classes. These figures were then added to the live crown biomass components. This provided crown biomass data for an additional 110 trees of the species of interest with dbh ≥ 5 cm. These data were merged with the data set described above for model development and calibration.

2.5. Model development

The primary objective of the data collection efforts was to develop predictive models of component biomass. To this end, trends in the means and variances of component biomass estimates were evaluated against differences in 4 variables commonly available in forest inventory data: tree species, dbh, total height, and crown length. Initial evaluations were made graphically and via correlations with and among potential predictors.

A number of linear and nonlinear model forms were considered to parameterize observed trends in stem and crown biomass. The basic candidate forms were drawn from [Zianis et al. \(2005\)](#) and from previous studies undertaken in western North America. Among those considered were log-linear power functions of the form

$$E[m] = \beta_0 x_1^{\beta_1} x_2^{\beta_2} \dots x_p^{\beta_p} = \exp\left(\ln\beta_0 + \sum_k \beta_k \ln(x_k)\right) \quad (1)$$

where m is a biomass component and the β are parameters of the expectation function to be estimated. To account for correlations among the predictor variables ($x_1, x_2, \dots, x_p$), power functions of these variables' principal components were also considered:

$$E[m] = \exp\left(\ln\beta_0 + \sum_j \beta_j c_{x_1 \dots x_p j}\right)$$

where the $c_{x_1 \dots x_p j}$ are principal components of $\ln(x_1), \ln(x_2), \dots, \ln(x_p)$ obtained after centering (but not scaling) these predictors. Owing to the log-linear form of (1), the latter equation can also be written as

$$E[m] = \tilde{\beta}_0 \exp\left(\sum_j \beta_j \left[\sum_k q_{j,k} \ln(x_k)\right]\right) = \tilde{\beta}_0 \prod_k x_k^{\left(\sum_j \beta_j q_{j,k}\right)} \quad (2)$$

where $q_{j,k}$ is the loading of $\ln(x_k)$ on principal component j and $\tilde{\beta}_0$ is a function of β_0 and the predictor variable means (on the logarithmic scale). Thus, the principal components formulation can be re-expressed as a power function of the original p predictors, but with a reduced number of parameters. In particular, if a single principal component of two or more of the original predictors is included, then (2) requires the estimation of only two parameters (β_0 and β_1).

To rank the candidate model forms, each was separately fit to the stemwood (m_w), stembark (m_b), live branch (m_l), dead branch (m_d), and foliage (m_f) biomass components using replicated 5-fold cross validation. In this procedure, a single replication consisted of a random partition of the set of available component biomass data for a given species into 5 folds. Each fold was withheld from one analysis and its biomass components were estimated from models calibrated using the other 4 folds. Model calibration was based on a Gaussian likelihood with residual variance proportional to an estimated power function of dbh. Differences between observed and estimated component biomass in the withheld fold were then summarized according to a Gaussian likelihood. This process was replicated 50 times allowing alternative model forms to be ranked for each species and component according to mean cross-validated likelihood (CVL; [Stone, 1977](#)). Alternative model forms employing the original predictors as well as models of the form (2) utilizing one or two principal components of dbh and height (or of dbh, height, and live crown length) were considered for all species and components.

Disaggregative functions of the form described by [Dong et al. \(2015\)](#) were used to describe the distribution of live branch mass across the three fuel time-lag classes. Specifically, the expected mass of live branches in a given combination of fuel classes was parameterized as

$$E[\delta_1 m_{1h} + \delta_{10} m_{10h} + \delta_{100+} m_{100+}] = \frac{\delta_1 \alpha_1 d^{\alpha_2} + \delta_{10} \alpha_3 d^{\alpha_4} + \delta_{100+}}{\alpha_1 d^{\alpha_2} + \alpha_3 d^{\alpha_4} + 1} \times E[m_l] \quad (3)$$

where δ_1, δ_{10} , and δ_{100+} are fixed binary indicator variables for the 1 h, 10 h, and 100+ hour class sizes respectively, and the α_i are species-specific coefficients to be estimated. This approach allows the proportions of live branch mass in each class to vary as a function of tree dbh. Tree characteristics other than (and in addition to) dbh were also evaluated in this context. However, in many cases dbh-based functions were superior and more complex functions could not be constrained to ensure that component biomass predictions were monotonically increasing with tree size.

Species-specific systems of biomass equations were obtained by combining models for each of m_f, m_l, m_d, m_b , and m_w with the live branch subcomponent function (3). Unrestricted systems were initially fit using the component equations with lowest CVLs. Constrained systems were also fit using only equations that included dbh and height, or that only included dbh. In these cases, systems were obtained by selecting component equations with lowest CVLs among those meeting the constraints. Parameters of the systems were estimated simultaneously from the full set of observed data using the Gaussian maximum likelihood approach described in [Affleck and Diéguez-Aranda \(2016\)](#) and implemented in the nlme package ([Pinheiro et al., 2018](#), ver. 3.1-131.1) in R ([R Core Team, 2018](#)). Unequal variances across and within components were modeled as power functions of dbh (i.e., as $\text{var}[m] = \theta_1^2 \text{dbh}^{2\theta_2}$ where θ_1 and θ_2 vary by species and component). Species-specific unstructured correlation matrices were used to characterize associations among components measured on the same trees,

but observations from distinct trees were treated as independent. Residuals for each component in a system were examined, and alternative model forms for individual components were replaced if systematic residual structure was observed. Depending on the CVL-based rankings of alternative equation forms, this replacement could amount to the elimination of an individual predictor (one of dbh, height, or live crown ratio; or a principal component of these) or to a change in the equation form itself. Also, alternative forms were evaluated if parameter estimates of a given component model were within 2 estimated standard errors of 0.

2.6. Model evaluation & comparison

Overall and size-related deviations of fitted biomass equations for particular species and components were evaluated using residual plots. Within-stand and within-plot correlations among Pearson residuals were also evaluated using linear mixed models with Gaussian random effects for stands and plots within stands. The importance of these correlations were judged according to differences in AIC (Akaike, 1974) between models with and without random effects. Overall system performance in terms of variability accounted for was summarized using a heteroscedasticity-corrected pseudo- R^2 statistic (see Parresol, 2001; Steele and Torrie, 1980, p. 270):

$$\text{pseudo-}R^2 = 1 - \frac{\sum \hat{\sigma}_i^{-2} (m_i - \hat{m}_i)^2}{\sum \hat{\sigma}_i^{-2} (m_i - \bar{m})^2} \quad (4)$$

where m_i is the observed component biomass for tree i , $\hat{m}_i$ denotes a component biomass estimate for that tree, $\hat{\sigma}_i^2$ is the estimated variance of the biomass component given the dbh of tree i , and $\bar{m} = \frac{\sum \hat{\sigma}_i^{-2} m_i}{\sum \hat{\sigma}_i^{-2}}$ is a variance-weighted mean of the component. The fitted equations' respective prediction errors were also summarized using mean absolute error (MAE)

$$\text{MAE} = 100 \times \frac{1}{n} \sum \frac{|m_i - \hat{m}_i|}{\hat{m}_i}, \quad (5)$$

a statistic that does not explicitly account for variance heterogeneity. The contributions of individual predictor variables to biomass component systems as a whole were evaluated for each species by evaluating using differences in system AIC.

3. Results

3.1. Data distribution

The destructively sampled data collected from 2009 to 2015 combined with those of Brown (1978) span a broad geographic range across the inland northwest (Fig. 1). The spatial distribution is unbalanced however, with data from the forested regions of south-central Montana, central Idaho, and northwestern Oregon being sparse or unavailable. The altitudinal profiles of each species' data set are shown in Fig. 2. Each species was observed over at least 1000 m of elevation, with the data for PICO spanning the largest range (≈ 1600 m).

The numbers of destructively sampled trees and their distribution across stands and plots are reported in Table 2. The PIEN and PICO tree samples were most dispersed in the sense that approximately 50% of these trees were the only representatives of the species in a plot, and approximately 20% were the only representatives of the species in a sampled stand. The PIPO sample was least dispersed: more than 80% of these trees were located in a plot with other sampled PIPO. Note also that not all biomass components were observed on all trees. In particular, the biomass data from Brown (1978) do not include stem mass information and not all data collected between 2009 and 2015 recorded stem bark mass separately from stemwood mass. For these reasons, there is overall more information available for crown mass components than for stem components.

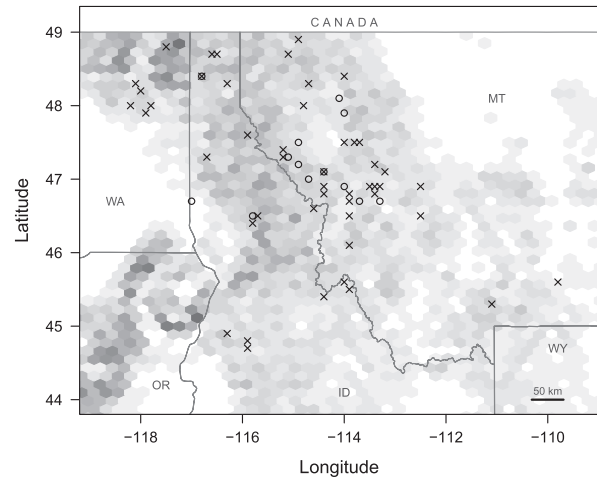


Fig. 1. Inland northwest region of the USA. Darker shading indicates higher relative aboveground tree biomass density (kg/ha) based on US Forest Inventory and Analysis data (USDA Forest Service, 2016). Crosses mark the geographic locations of sample trees collected for this study; circles denote approximate geographic locations of data collected by Brown (1978).

The size distributions of the sample trees are depicted in Fig. 3. Size class representations differed by species, but overall the trees ranged in dbh from 5 cm to over 1 m and from 3 m to 44 m in height. At the same time, linear correlations between dbh and height ranged from 0.88 to 0.92 across the 7 species. The PICO and LAOC samples covered a range of crown lengths within dbh and height classes (linear correlations of live crown length to height or to dbh below 0.8), but for other species, especially PIPO and PIEN, crown length was strongly associated dbh and height (correlations > 0.88).

3.2. Model selection

Excepting dead branch biomass, which exhibited substantial variation for all 7 species, component biomass estimates were strongly associated with tree basal area (Fig. S-1). On the logarithmic scale, m_b and m_w scaled approximately linearly with basal area, while m_f and m_h exhibited nonlinear associations (for most species). Also, variability in component biomass given tree basal area was homogeneous on this scale. The patterns supported the use of power functions of the form (1) in component model specifications, as well as dbh-related variance functions for modeling biomass on the original scale.

The performance of different candidate models as ranked by mean CVL is summarized for PSME in Fig. 4, and for all species in the supplementary materials (Fig. S-2). These figures break out candidate component equations according to the types of predictors utilized (i.e., dbh only vs. dbh and height vs. any of dbh, height, crown length) and by the number of parameters in the expectation function (all models were fit with the same type of variance function). In general, dbh-based models performed poorly relative to those that incorporated additional predictors. The only exception was the case of dead branch biomass where 3-parameter dbh-based equations had the lowest CVLs for PSME (Fig. 4), PIEN, and LAOC (Fig. S-2). For the other species, equations utilizing dbh and height (but not crown length) had better overall performance in estimating dead branch biomass. By contrast, foliage and live branch mass equations that utilized live crown length had the smallest CVLs for most species, including PSME. The cross-validation analyses also indicated that equations utilizing live crown length for m_b estimation were most effective for PSME and ABGR. However, for PSME and most other species, equations using only dbh and height performed best for m_w estimation.

Overall performance as indicated by mean CVL was used to guide

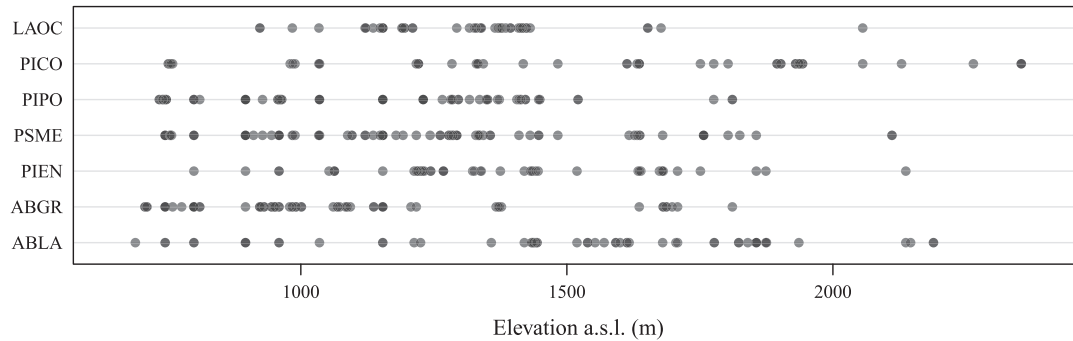


Fig. 2. Distribution of sample trees by species and elevation (jittered ± 25 m for clarity). Elevations for tree data collected by Brown (1978) are based on approximate geographic coordinates. Species are arranged from bottom to top in order of decreasing shade tolerance (see Baker, 1949; Lienard et al., 2015).

initial identification of equation forms for model system estimation. However, the best equation forms for individual components as determined by this metric were not necessarily selected in the end. This was in part because the CVL metric, like most other point estimates of model fit, only described overall or marginal performance. In selecting final model forms, residual plots were also screened and equation forms with poor conditional performance (e.g., conditional on dbh or crown ratio) were replaced. Also, several component equation forms were substituted or simplified after model system estimation indicated that a subset of parameter estimates were within 2 standard errors of 0.

The systems of equations drawing on dbh, height, and live crown length that were ultimately selected are shown in a simplified parametric form in Table 3. The native parameterization of these systems and of those using only equations drawing on dbh and height are detailed in the supplementary materials (Table S-1). All component equations utilized dbh with positive exponents averaging approximately 2.4 for live branches, 2.0 for foliage, and 1.75 for the other components. The selected systems all utilized height for dead branch biomass estimation and live crown length for estimation of foliage and live branch biomass. The crown length effects for foliage and live branch estimation were all positive (i.e., an increased live crown length was associated with higher mass). Height was also selected as a predictor of foliage and live branch biomass for some species, and was consistently associated with negative exponents in those cases. By contrast, height as a predictor of stembark or stemwood mass was always associated with positive exponents whereas crown length, if used, inversely affected mass estimates (longer live crowns associated with reduced bark or wood mass in ABGR and PSME). For most species, the principal components form of the log-linear power function in dbh and height was selected for stemwood mass estimation such that the pair of estimated exponents presented for these two variables in Table 3 are typically derived from a single estimand. An exception to this is PSME, for which the best performing system of equations drew on the first two principal components of $\ln d$, $\ln h$, and $\ln l$ for the estimation of m_b but only the first principal component of $\ln d$ and $\ln h$ for estimation of m_w

(Table S-1). Also, the system selected for ABGR utilized alternative equation forms for m_b and m_w and the original predictors.

Across all species, there was high variability in $m_{l_{100+}}$, both in absolute terms and as a proportion of overall live branch mass. As a result, the 100+ hour fuel class and the 10 h fuel class were combined and the disaggregation function simplified to

$$E[\delta_1 m_{l_1} + \delta_{10+} m_{l_{10+}}] = \frac{\delta_1 \alpha_1 d^{\alpha_2} + \delta_{10+}}{\alpha_1 d^{\alpha_2} + 1} \times E[m_l] \quad (6)$$

such that it discriminated only between live branch mass in the 1 h size class and mass in the 10+ hour size classes. Estimated coefficients for these disaggregation functions were similar across species except for PIPO which had very low levels of m_{l_1} owing to the fact that its branches rarely tapered below the 0.635 cm diameter threshold.

The estimated covariance parameters for the model systems of Table 3 are shown in Table 4. The estimates show that across all species and components, variance in biomass increased as a function of dbh (i.e., $\hat{\sigma}_2 > 0$). Also, cross-correlations among foliage mass, live branch mass in the 1 h class, and live branch mass in the 10+ hour class on the same trees were consistently positive and often exceeded 0.5.

Residual plots for key biomass components and subtotals of biomass components of the model systems in Table 3 are provided in the supplementary materials (Fig. S-3). These plots suggested that the fitted systems captured the overall multivariate trends in biomass allometries and that the dbh-based heteroscedasticity models appropriately stabilized variances. However, there were a number of small trees that deviated from the fitted equations substantially on the variance-stabilized scaled. These trees contributed to heavier than anticipated tails under the Gaussian model for some species and some components (most commonly the dead branch mass component). Linear models applied to the Pearson residuals identified significant differences (at the 10% level) across crown classes (i.e., dominant, codominant, etc.) in only 5 of 35 cases (7 species $\times$ 5 primary biomass components). Similarly, applying linear mixed models to these residuals, random effects representing stand and plot-within-stand heterogeneity improved model-

Table 2
Numbers of stands, plots, trees, and component biomass observations by species.

Species	No. stands	No. plots	No. trees	Biomass component							
				Dead br.	Fol.	1 h	Live br.	Stembark	Stemwood	Stem	Abovestump
ABLA	24	36	57	52	57	52	57	44	44	44	52
ABGR	26	40	73	59	71	59	71	47	47	48	59
PIEN	20	40	49	40	47	39	47	42	42	43	40
PSME	37	54	92	88	92	71	92	42	42	62	88
PIPO	23	40	85	81	84	52	84	22	22	51	81
PICO	23	37	56	54	56	42	56	42	42	55	54
LAOC	18	41	58	44	58	32	58	36	36	51	44

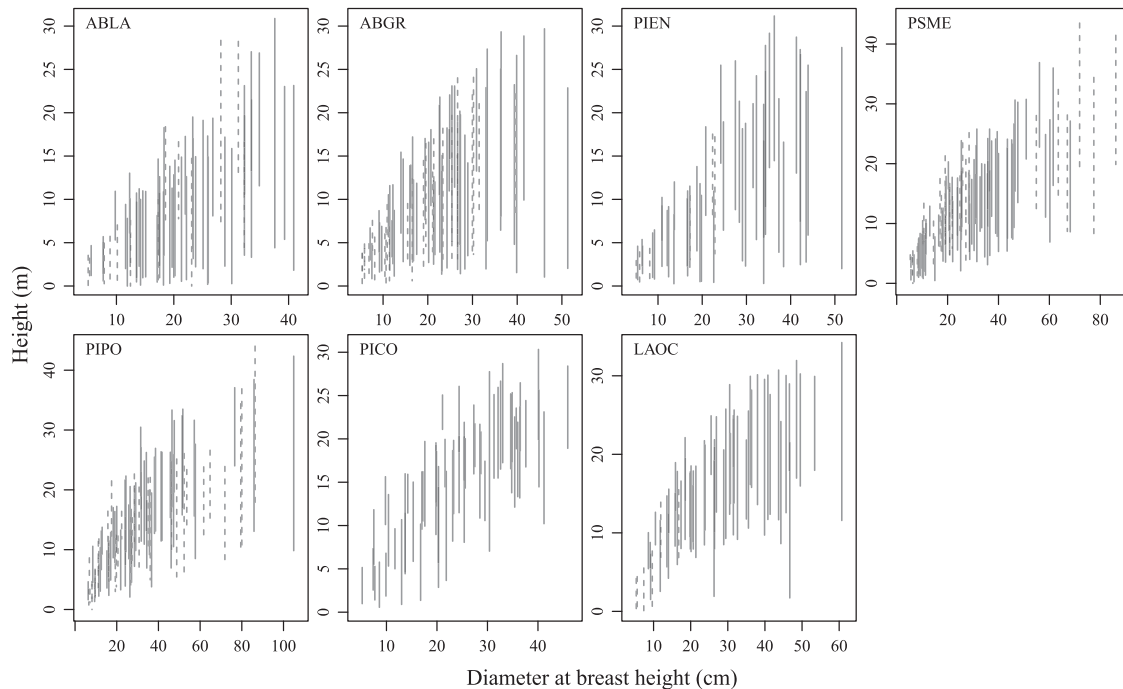


Fig. 3. Size distribution of sample trees by species: vertical line segments run from crown base height to total tree height. Trees collected by Brown (1978) are shown with dashed line segments.

data agreement ($\Delta AIC > 5$) in only 5 cases: for two LAOC components (live branches and foliage), two PIEN components (dead branches and whole crown), and PSME stembark. For all other species and components the system residuals did not exhibit appreciable cluster correlations within plots or stands.

Goodness-of-fit of the selected equation systems of Table 3 are shown in Figs. 5 and 6. In general, estimation equations for total above-stump, total stem, and stemwood biomass had pseudo- R^2 values above 0.95 and MAEs of approximately 10%. From there, variance accounted for declined, and MAEs increased, across the equations for m_b , m_c , m_l , and m_f . The equations for dead branch biomass accounted for only 41–67% of variability on the normalized scale, with mean prediction errors on the original scale ranging as high as 81%. The same trends were observed for equation systems utilizing live crown length

and for those using only dbh and height. However, on the basis of AIC, equation systems developed with access to live crown length as a predictor proved superior to systems without access to live crown length for all species but LAOC (Fig. 7). Also, systems using dbh and height proved superior to those using only dbh for all species.

4. Discussion

The biomass models developed in this study are based on 470 destructively sampled trees sourced from 84 stands in the inland north-west. Despite its extent, the sample was unequally distributed across the region (Fig. 1), across tree sizes (Fig. 3), and across species (Table 2). These imbalances reflect practical challenges associated with obtaining felled tree data. The necessarily destructive nature of the measurements

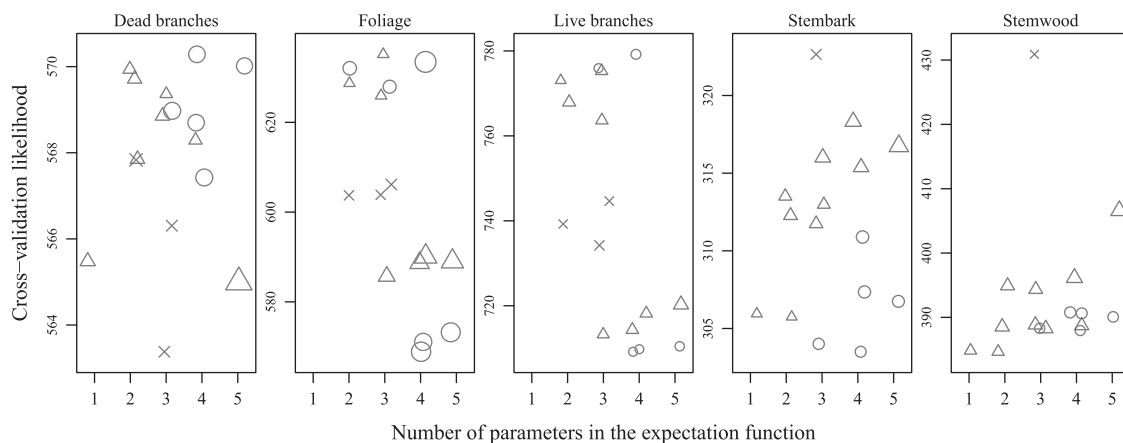


Fig. 4. Mean cross-validated likelihoods (CVL) of fitted component equations for PSME; only equations with mean CVLs in the lower 80% of the component range are represented. Equations using dbh only are indicated by a cross; triangles denote equations using dbh and height; circles denote equations using crown length as well as dbh and height. Symbols are sized according to the standard deviation in CVLs obtained over 50 replications.

Table 3

Calibrated systems of tree component biomass (kg) equations by species. Mean functions were selected from among those using dbh (cm), height (m), and crown length (m) as predictors. Also tabulated are the numbers of parameters in the mean functions.

Species	Component	Mean function (kg)	No. param.
ABLA	Dead branches	$\hat{m}_d$ 0.0006919 $d^2 h$	1
	Foliage	$\hat{m}_f$ 0.02334 $d^{1.669} e^{-1.927 l/h}$	3
	Live branches	$\hat{m}_l$ 2.108 + 0.01162 $d^{2.489} h^{-1.255} l^{1.362}$	5
	1 h fuels	$\hat{m}_{l1}$ $\hat{m}_l / (1 + 0.9260 d^{0.1729})$	2
	Stembark	$\hat{m}_b$ 0.003515 $(d^2 h)^{0.9177}$	2
	Stemwood	$\hat{m}_w$ 0.01259 $d^{1.485} h^{1.506}$	2
ABGR	Dead branches	$\hat{m}_d$ 0.002828 $(d^2 h)^{0.8606}$	2
	Foliage	$\hat{m}_f$ 0.006025 $d^{1.748} e^{3.563 l/h}$	3
	Live branches	$\hat{m}_l$ 0.09946 $d^{2.173} h^{-1.718} l^{1.542}$	4
	1 h fuels	$\hat{m}_{l1}$ $\hat{m}_l / (1 + 0.9250 d^{0.3047})$	2
	Stembark	$\hat{m}_b$ 0.002800 $d^{2.780} (l/h)^{-0.4696}$	3
	Stemwood	$\hat{m}_w$ 0.01785 $d^{1.828} h^{1.181} e^{-0.5256 l/h}$	4
PIEN	Dead branches	$\hat{m}_d$ 0.0007322 $d^2 h$	1
	Foliage	$\hat{m}_f$ 0.007975 $d^{2.010} e^{2.267 l/h}$	3
	Live branches	$\hat{m}_l$ 0.009508 $d^{2.230} l^{1.638 l/h}$	3
	1 h fuels	$\hat{m}_{l1}$ $\hat{m}_l / (1 + 0.6725 d^{0.2933})$	2
	Stembark	$\hat{m}_b$ 0.005327 $d^{1.322} h^{1.296}$	2
	Stemwood	$\hat{m}_w$ 0.009904 $d^{1.410} h^{1.681}$	3
PSME	Dead branches	$\hat{m}_d$ 0.0009405 $d^2 h$	1
	Foliage	$\hat{m}_f$ 0.03072 $d^{2.240} h^{-0.6511} e^{1.412 l/h}$	4
	Live branches	$\hat{m}_l$ 0.02340 $d^{2.700} h^{-0.7130} e^{1.020 l/h}$	4
	1 h fuels	$\hat{m}_{l1}$ $\hat{m}_l / (1 + 0.4468 d^{0.6396})$	2
	Stembark	$\hat{m}_b$ 0.002090 $d^{1.673} h^{1.688} l^{-0.4072}$	3
	Stemwood	$\hat{m}_w$ 0.01053 $d^{1.714} h^{1.390}$	2
PIPO	Dead branches	$\hat{m}_d$ 0.003204 $d^{2.698} e^{-0.04444 h}$	3
	Foliage	$\hat{m}_f$ 0.02924 $d^{2.350} h^{-0.7552} l^{0.3080}$	3
	Live branches	$\hat{m}_l$ 0.01837 $d^{3.016} h^{-1.099} l^{0.4196}$	4
	1 h fuels	$\hat{m}_{l1}$ $\hat{m}_l / (1 + 0.6574 d^{1.279})$	2
	Stembark	$\hat{m}_b$ 0.006382 $(d^2 h)^{0.8865}$	2
	Stemwood	$\hat{m}_w$ 0.009907 $d^{1.658} h^{1.410}$	2
PICO	Dead branches	$\hat{m}_d$ 0.001504 $d^{2.698} e^{-0.04444 h}$	2
	Foliage	$\hat{m}_f$ 0.002336 $d^{2.286} e^{2.514 l/h}$	3
	Live branches	$\hat{m}_l$ 0.003366 $d^{2.529} e^{1.820 l/h}$	3
	1 h fuels	$\hat{m}_{l1}$ $\hat{m}_l / (1 + 0.5379 d^{0.5277})$	2
	Stembark	$\hat{m}_b$ 0.005704 $(d^2 h)^{0.8207}$	2
	Stemwood	$\hat{m}_w$ 0.008407 $d^{1.754} h^{1.451}$	2
LAOC	Dead branches	$\hat{m}_d$ 0.02487 $(d^2 h)^{0.4831}$	2
	Foliage	$\hat{m}_f$ 0.03913 $d^{1.611} (l/h)^{0.7428}$	4
	Live branches	$\hat{m}_l$ 0.05256 $d^{2.034} (l/h)^{0.7147}$	3
	1 h fuels	$\hat{m}_{l1}$ $\hat{m}_l / (1 + 0.3244 d^{0.6291})$	2
	Stembark	$\hat{m}_b$ 0.001910 $d^{1.636} h^{1.332}$	2
	Stemwood	$\hat{m}_w$ 0.008532 $d^{1.753} h^{1.428}$	2

hampered access to trees and stands, and both field and laboratory measurements were resource intensive. The elevation gradient spanned by the sample plots (Fig. 2) is a function of the mountainous and complex terrain found across the inland northwest. A broad range of elevation was sought for each target species owing to the fact that each grows in diverse forest habitat types (see, e.g., Pfister et al., 1977) that, in turn, are often differentiated along elevation gradients.

The imbalance in biomass observations by component (Table 2) is partly a function of the differential sampling costs of stem and crown elements. However, the greatest part of this imbalance is attributable to the inclusion of Brown (1978) crown biomass data. Combining those crown-only data with our felled tree observations was deemed advantageous in part because Brown captured some very large trees (e.g., PSME and PIPO with dbh > 75 cm) and in part because crown components are both more time-consuming to measure and inherently more

variable. One issue associated with the combination of data sets is the fact that Brown determined the base of the live crown slightly differently, as “the position on the bole where a full crown could exist if lower branches were moved up to fill in open spaces” (p. 5). Nonetheless, no differences were observed between the two data sets in the distribution of live crown lengths by dbh, and no significant differences between data sets were observed in the foliage, live branch, or whole crown residuals of the fitted equation systems of Table 3.

With the exception of dead branch mass, component biomass scaled approximately linearly with tree dbh (Fig. S-1). Curvature in those trends is here attributed to variations in height:dbh ratios and/or in crown lengths. For a given dbh, variations may reflect differences in site quality (Case and Hall, 2008) or stand density, and lead directly to differences in stem volume. Equations utilizing both height and dbh were selected for stem mass components for all species (Table S-1), and the fact that the partial effect of height was positive suggests that its impact on stem volume was a dominant effect. Indeed, for most species, the best performing equations for stemwood mass utilized only projections of log-dbh and log-height onto these two variables’ first principal component, corresponding to the utilization of a single metric of overall size. Equations drawing on height were also commonly selected for live crown mass components (m_f and m_l). However, the partial association between height and these components was consistently negative (whether or not live crown length was utilized), in agreement with the mechanistic explanation of Zeide (1987).

Although live crown length was incorporated into biomass equations for some stem components for some species, the most consistent and important contribution of this predictor was to the live crown components (Figs. 4 and S-2). This predictor was positively associated with foliage and live branch mass across all species, in agreement with previous results from the region and with allometric scaling theory (Brown, 1978; Evert, 1985; Mäkelä and Valentine, 2006). The increases in system AIC associated with dropping live crown length was appreciable for all species except LAOC (Fig. 7) and it is apparent from Fig. 6 that the accuracies of the live crown component equations are degraded more than are those of the stem component equations. The consistently positive within-tree cross-correlations among foliage and live branch biomass components (Table 4) is likely attributable at least as much to the sampling methods employed (each of these was estimated from the same sample branches) as is to intrinsic associations among these attributes. Regardless, the cross-correlation is an important feature of the data to recognize and capitalize upon to improve parameter estimation.

Within species and even within size classes, there was considerable variation in the observed levels and proportions of live branch mass in the >100-h time lag class. This variation is partly attributable to the branch sampling procedure. RBS was not constrained to capture branches of all size classes and, despite its probability-proportional-to-size nature, can result in selection of sample branches with basal diameters strictly below 2.54 cm even on larger trees where large branches are present. Additionally, owing to the fact that the tree tip above the threshold upper diameter of 5 cm was considered part of the crown, small trees (dbh < 8 cm) could have a considerable proportion of their live branch mass in the >100-h time lag class. Aggregation of 10-h and >100-h branch masses allowed for modeling size class differentiation at the fine vs. coarse level, though did result in a reduction in resolution. Ideally, the diminution of live branch mass by cross-sectional diameter would be modeled continuously in a manner analogous to that used by Weiskittel and Maguire (2006) to model branch surface area. However, the requisite data (sequential branch diameters, branch lengths) were not collected in this study.

The systems of equations presented in Table 3 provide a means of accurately estimating abovestem biomass, stem biomass, and stemwood biomass across the region (Fig. 6). Naturally, the accuracy of total stem biomass and total abovestem biomass estimates follows from the fact that stemwood biomass comprises an appreciable proportion of these totals and is itself well described on the basis of dbh and height

Table 4

Covariance parameters of the biomass equation systems; $\hat{\sigma}_0$ and $\hat{\sigma}_1$ define the fitted variance function for each component and the triangular matrices provide estimated cross-correlations among components of the same tree.

Species	Component	$\hat{\sigma}_1$	$\hat{\sigma}_2$	m_f	m_{l1}	m_{l10+}	m_b	m_w
ABLA	Dead branches	0.005995	2.0620	0.0101	0.1425	0.0309	0.3094	0.1328
	Foliage	0.031259	1.8006		0.3159	0.4354	−0.0726	−0.1915
	1 h fuels	0.009288	1.9551			0.3894	0.2127	0.1225
	10 + hour fuels	0.003425	2.4131				0.1535	0.0676
	Stembark	0.000876	2.6576					0.3202
	Stemwood	0.018606	1.9602					
ABGR	Dead branches	0.003248	2.3250	0.2441	0.0207	0.2029	−0.0829	−0.3020
	Foliage	0.003370	2.6021		0.6528	0.7073	−0.2049	−0.2431
	1 h fuels	0.003853	2.3841			0.3824	−0.2434	−0.2868
	10 + hour fuels	0.015259	2.1303				−0.4603	−0.1791
	Stembark	0.002768	2.3186					0.3340
	Stemwood	0.003878	2.6832					
PIEN	Dead branches	0.002287	2.3926	−0.1887	−0.0743	0.1282	−0.0830	0.2775
	Foliage	0.012026	2.1387		0.6874	0.5711	−0.0505	0.4499
	1 h fuels	0.005531	2.2345			0.6597	0.0648	0.4030
	10 + hour fuels	0.018626	1.8921				0.0341	0.2079
	Stembark	0.001625	2.3598					−0.0061
	Stemwood	0.012572	2.1964					
PSME	Dead branches	0.001311	2.6828	−0.0350	−0.0187	0.0641	0.0334	−0.0054
	Foliage	0.044025	1.5037		0.7943	0.5985	0.0652	0.3970
	1 h fuels	0.025131	1.4775			0.6372	−0.1686	0.1807
	10 + hour fuels	0.019464	1.9389				0.1737	0.4475
	Stembark	0.000547	2.9215					0.1872
	Stemwood	0.003571	2.6303					
PIPO	Dead branches	0.004497	2.2272	−0.1449	0.1261	0.1740	0.6598	−0.0656
	Foliage	0.011950	1.9694		0.3333	0.5840	−0.2900	0.4523
	1 h fuels	0.003936	1.6285			0.1854	−0.0212	0.1092
	10 + hour fuels	0.005189	2.4091				−0.0416	0.2337
	Stembark	0.003278	2.4000					0.2458
	Stemwood	0.000336	3.3969					
PICO	Dead branches	0.002173	2.4618	0.0223	0.4651	0.2142	−0.0367	−0.0764
	Foliage	0.006166	2.1222		0.1827	0.4020	0.0617	0.0742
	1 h fuels	0.159779	0.9401			0.6070	0.0962	0.0592
	10 + hour fuels	0.043935	1.6058				0.0514	0.0358
	Stembark	0.012898	1.6845					−0.0253
	Stemwood	0.032606	2.0693					
LAOC	Dead branches	0.032341	1.3557	−0.2164	0.1155	−0.1938	0.2841	0.2458
	Foliage	0.020332	1.4119		0.7921	0.6710	−0.3020	0.2149
	1 h fuels	0.011654	1.6573			0.6691	−0.2994	0.1923
	10 + hour fuels	0.007942	2.1417				−0.0578	0.1379
	Stembark	0.000120	3.3176					−0.1292
	Stemwood	0.005510	2.6018					

(Fig. 5). These same two predictors capture much of the variation in stembark biomass, but the estimation equations have reduced accuracy relative to stemwood biomass. We attribute this reduced accuracy to within-species and within-tree variations in bark density and

morphology. Crown components are estimated with the lowest accuracy despite larger sample sizes (Table 2), reflecting the relative variability of crown architecture given dbh, height, and crown length. The estimation errors associated with the crown components are large

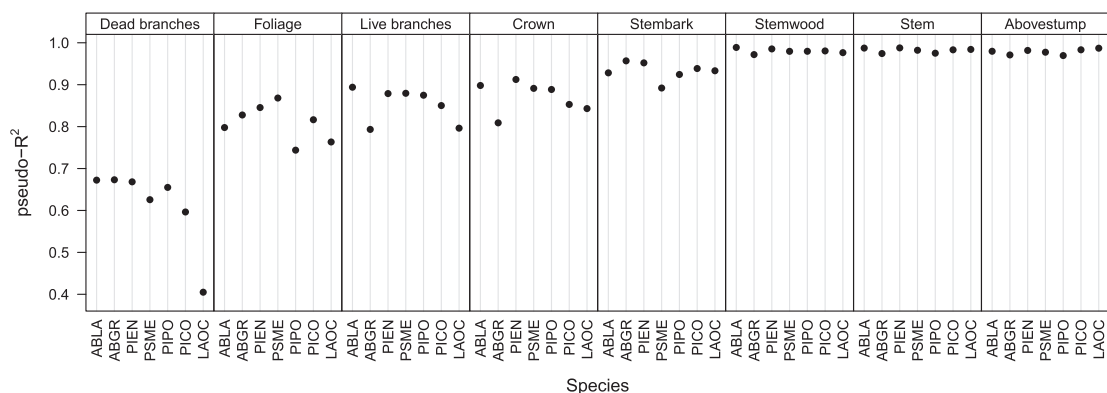


Fig. 5. Goodness-of-fit of select component submodels by species; model systems are detailed in Table 3.

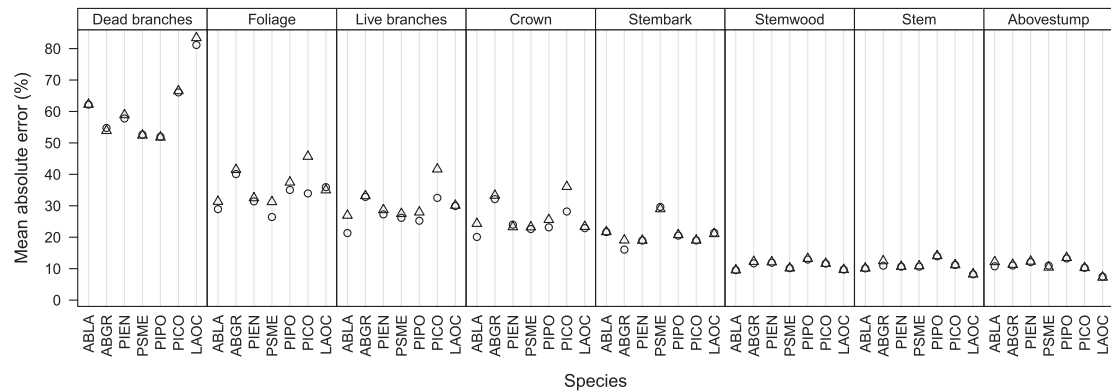


Fig. 6. Prediction error summaries of select component models by species; circles denote component equations from model systems detailed in Table 3 and triangles denote component equations from model systems utilizing only dbh and height (see Table S-1).

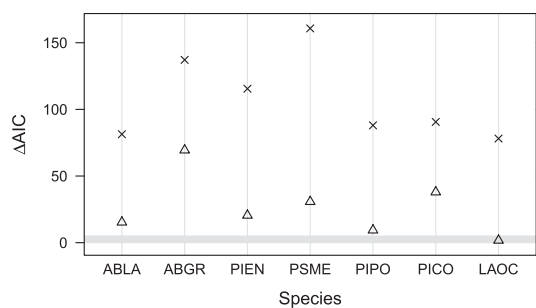


Fig. 7. Differences in AIC between selected model systems accessing dbh, height, and crown length as predictors vs. systems employing dbh and height (triangles) or vs. systems employing dbh only (crosses).

(~30% for live crown components, >50% for dead branches), but are in line with those reported in other regional studies (e.g., see Brown, 1978).

In conclusion, the systems of biomass equations developed in this study draw from the largest and most widely distributed felled-tree database yet compiled for the region. They condition on standard forest inventory metrics, and provide a compatible basis for component-level estimation of biomass. Unlike previously developed systems for the region they include equations for stem components while also capturing differences in biomass allometries across species, height:dbh ratios, and crown lengths. As such, we believe they offer an improved basis for biomass estimation for the most ubiquitous conifer species in the inland northwest. Algorithms for estimating component and total tree biomass with standard errors are detailed in the supplementary materials (p. S21).

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.foreco.2018.09.009>.

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